



Proton NMR Spectroscopy

Video Series 2

Video 2: NMR Solvents





Video Series Goals

Video 1: NMR Spectra of Mixtures

- Diagnostic NMR signals
- Calculating Percent Composition of a Mixture

Video 2: NMR Solvents (**Current**)

- Deuterated Solvents
- Solvent Signals in NMR

Video 3: Splitting Pattern Complexity

- Extension of “n+1” Rule

Video 4: Long Range Coupling

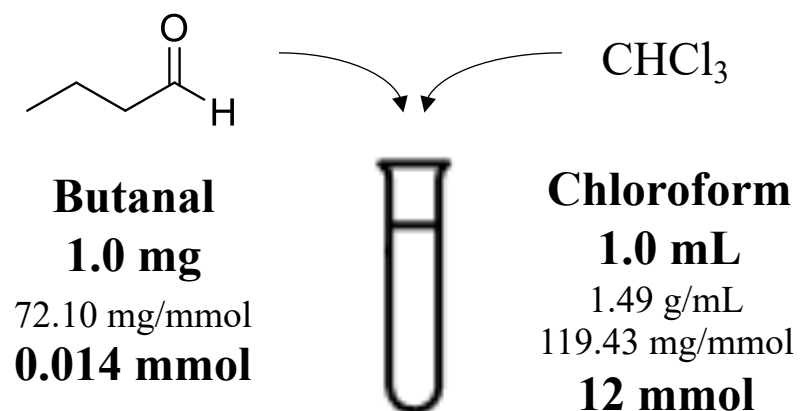
- 4-Bond (4J) Coupling

Video 5: Diastereotopic Protons

- 2-Bond (2J) Coupling

NMR Solvents

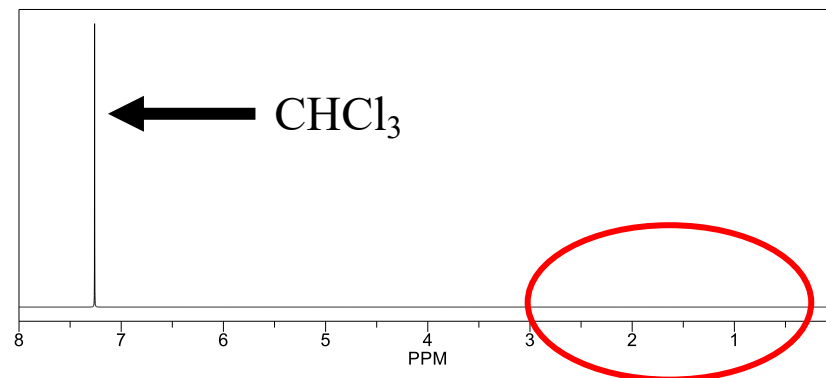
^1H NMR experiments are run at low concentration ($\sim 1\text{-}5\text{ mg/mL}$)



Need an NMR solvent with no protons!

1 mg of Butanal in 1 mL of chloroform:

- There are **860 chloroform molecules** for every **1 butanal molecule!**
- Will not see any butanal signals in the NMR





Deuterated Solvents

^1H NMR experiments are run at low concentration ($\sim 1\text{-}5$ mg/mL)

^1H NMR experiments must be run in **deuterated solvents**

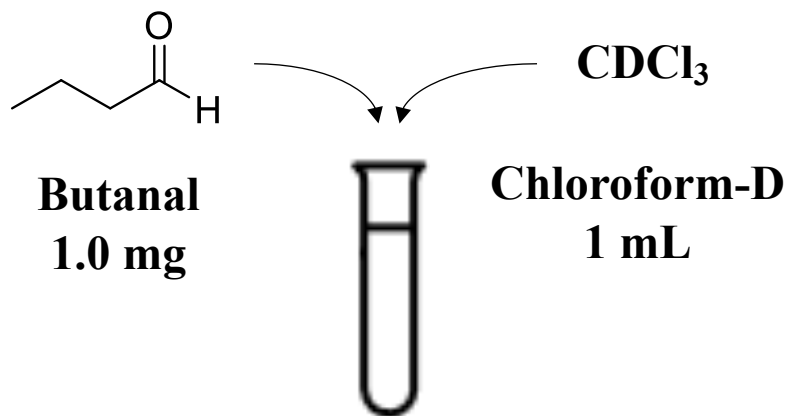
- ^2H (D, deuterium) is NMR active, but is **not visible in a ^1H NMR spectrum**
- Deuterated solvents are typically $> 99.8\%$ D and $< 0.2\%$ H
 - CDCl_3 will always have some CHCl_3 impurities (Residual solvent peak)

Deuterated Solvents

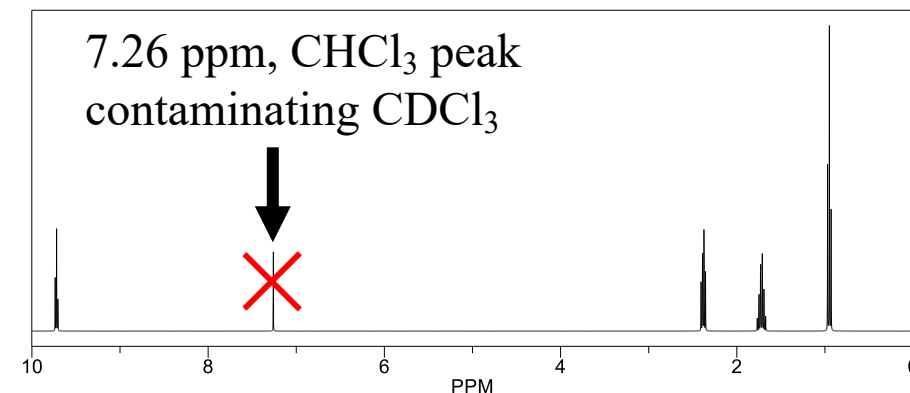
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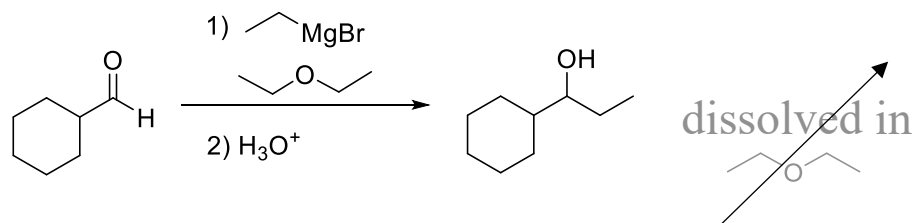


^1H NMR of butanal in CDCl_3



Solvent contamination in NMR spectra

Grignard reactions are often run in diethyl ether

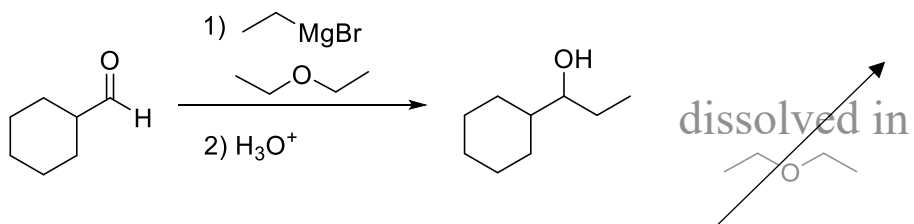


Remove with:

- Heat
- Vacuum

Solvent contamination in NMR spectra

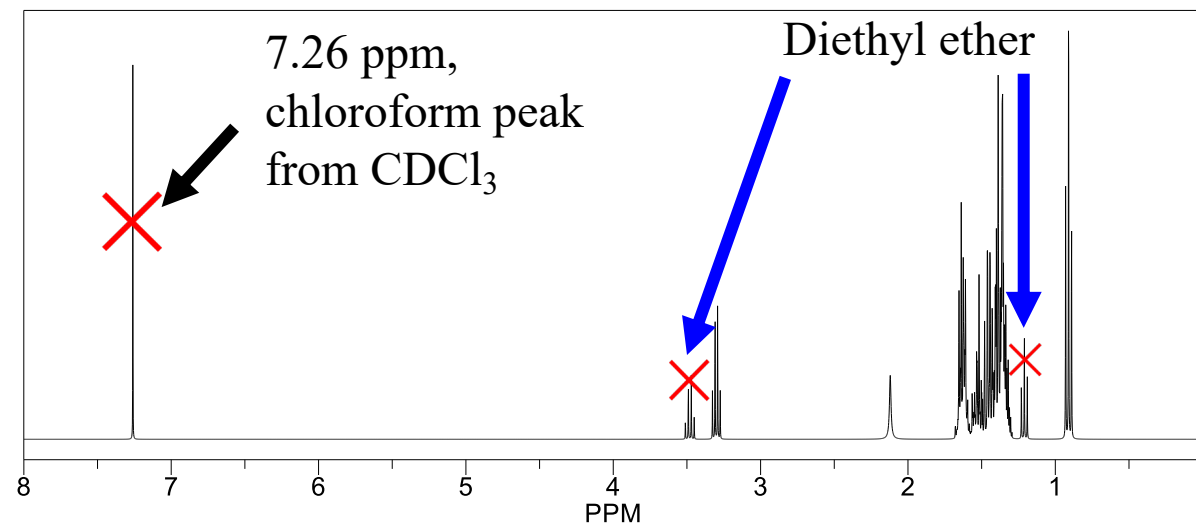
Grignard reactions are often run in diethyl ether



Remove with:

- Heat
- Vacuum

^1H NMR of product in CDCl_3



Solvent contamination in NMR spectra

NMR Chemical Shifts of Common Laboratory Solvents as Trace Impurities

Hugo E. Gottlieb,* Vadim Kotlyar, and
Abraham Nudelman*

Department of Chemistry, Bar-Ilan University,
Ramat-Gan 52900, Israel

Received June 27, 1997

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- Residual peak is due to undeuterated NMR solvent impurities
- Chemical shift depends on the NMR solvent used

Appendix II

Table 1. ^1H NMR Data

	proton	mult	CDCl_3	$(\text{CD}_3)_2\text{CO}$	$(\text{CD}_3)_2\text{SO}$	C_6D_6	CD_3CN	CD_3OD	D_2O
solvent residual peak			7.26	2.05	2.50	7.16	1.94	3.31	4.79
H_2O		s	1.56	2.84 ^a	3.33 ^a	0.40	2.13	4.87	
acetic acid	CH_3	s	2.10	1.96	1.91	1.55	1.96	1.99	2.08
acetone	CH_3	s	2.17	2.09	2.09	1.55	2.08	2.15	2.22
acetonitrile	CH_3	s	2.10	2.05	2.07	1.55	1.96	2.03	2.06
benzene	CH	s	7.36	7.36	7.37	7.15	7.37	7.33	
<i>tert</i> -butyl alcohol	CH_3	s	1.28	1.18	1.11	1.05	1.16	1.40	1.24
	OH^c	s			4.19	1.55	2.18		
<i>tert</i> -butyl methyl ether	CCH_3	s	1.19	1.13	1.11	1.07	1.14	1.15	1.21
	OCH_3	s	3.22	3.13	3.08	3.04	3.13	3.20	3.22
BHT ^b	ArH	s	6.98	6.96	6.87	7.05	6.97	6.92	
	OH^c	s	5.01		6.65	4.79	5.20		
	ArCH_3	s	2.27	2.22	2.18	2.24	2.22	2.21	
	$\text{ArC}(\text{CH}_3)_3$	s	1.43	1.41	1.36	1.38	1.39	1.40	
chloroform	CH	s	7.26	8.02	8.32	6.15	7.58	7.90	
cyclohexane	CH_2	s	1.43	1.43	1.40	1.40	1.44	1.45	
1,2-dichloroethane	CH_2	s	3.73	3.87	3.90	2.90	3.81	3.78	
dichloromethane	CH_2	s	5.30	5.63	5.76	4.27	5.44	5.49	
diethyl ether	CH_3	t, 7	1.21	1.11	1.09	1.11	1.12	1.18	1.17
	CH_2	q, 7	3.48	3.41	3.38	3.26	3.42	3.49	3.56

Solvent contamination in NMR spectra

¹H NMR of product in CDCl₃

NMR Chemical Shifts of Common Laboratory Solvents as Trace Impurities

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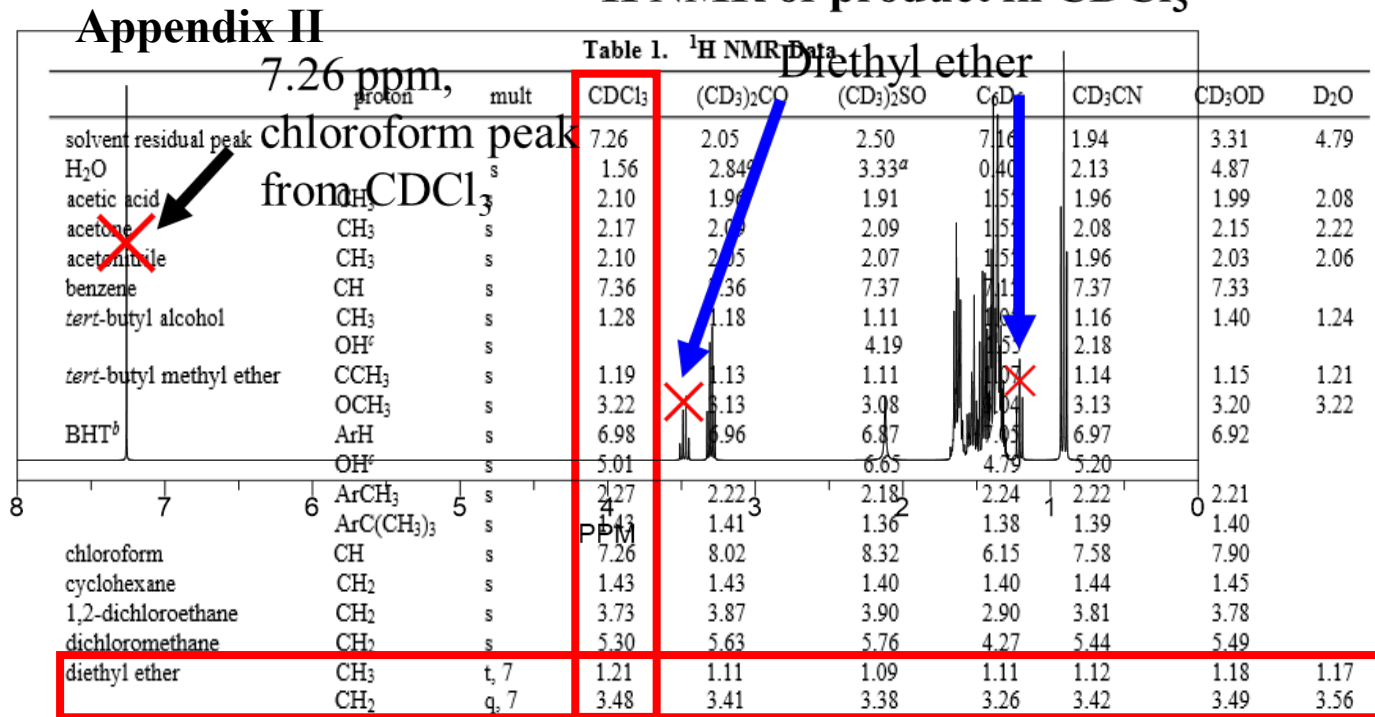
- Residual peak is due to undeuterated NMR solvent impurities
- Chemical shift depends on the NMR solvent used

Appendix H

Table 1. ¹H NMR Data

7.26 ppm, proton, mult, chloroform peak from CDCl₃

Diethyl ether



	proton	mult	CDCl ₃	(CD ₃) ₂ CO	(CD ₃) ₂ SO	C ₆ D ₆	CD ₃ CN	CD ₃ OD	D ₂ O
solvent residual peak			7.26	2.05	2.50	7.16	1.94	3.31	4.79
H ₂ O		s	1.56	2.84 ^a	3.33 ^a	0.40	2.13	4.87	
acetic acid	CH ₃	s	2.10	1.96	1.91	1.51	1.96	1.99	2.08
acetone	CH ₃	s	2.17	2.09	2.09	1.51	2.08	2.15	2.22
acetonitrile	CH ₃	s	2.10	2.05	2.07	1.51	1.96	2.03	2.06
benzene	CH	s	7.36	7.36	7.37	7.11	7.37	7.33	
tert-butyl alcohol	CH ₃	s	1.28	1.18	1.11	1.16	1.16	1.40	1.24
	OH ^c	s			4.19	2.18			
tert-butyl methyl ether	CCH ₃	s	1.19	1.13	1.11	1.14	1.14	1.15	1.21
	OCH ₃	s	3.22	3.13	3.08	3.13	3.13	3.20	3.22
BHT ^b	ArH	s	6.98	6.96	6.87	6.93	6.97	6.92	
	OH ^c	s	5.01		6.65	4.79	5.20		
	ArCH ₃	s	2.27	2.22	2.18	2.24	2.22	2.21	
	ArC(CH ₃) ₃	s	1.43	1.41	1.36	1.38	1.39	1.40	
chloroform	CH	s	7.26	8.02	8.32	6.15	7.58	7.90	
cyclohexane	CH ₂	s	1.43	1.43	1.40	1.40	1.44	1.45	
1,2-dichloroethane	CH ₂	s	3.73	3.87	3.90	2.90	3.81	3.78	
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diethyl ether	CH ₃	t, 7	1.21	1.11	1.09	1.11	1.12	1.18	1.17
	CH ₂	q, 7	3.48	3.41	3.38	3.26	3.42	3.49	3.56



Video Summary

Must use deuterated solvents to make NMR samples

- Residual solvent peaks are due to undeuterated impurities in the NMR solvent (usually $< 0.2\%$ ^1H)

Solvents from running reactions and purifying compounds are common NMR impurities

- “NMR Chemical Shifts of Common Laboratory Solvents as Trace Impurities” is a very useful resource when examining the NMR spectra of synthesis products

Identification of solvent impurities is an important first step when analyzing NMR data

Next video: Complex Splitting Patterns